# MGSC 706: Management Research Statistics

Lecture Notes

PhD Core Course in Management Research Statistics

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These notes cover the course lecture material for the PhD core course in management research statistics.

# Contents

MGSC 706 is a core doctoral course in management research statistics. Its scope proceeds from probability to sampling, statistical inference, hypothesis testing, regression, and model building. Although formulas are essential, the course emphasizes business applications and interpretation. Our plan is to demonstrate Python in a cloud environment, while allowing students to use other statistical platforms such as R or SPSS when appropriate.

The pedagogical principle is learning by doing. Manual calculations are useful for understanding definitions, but real research datasets require software. The deeper goal is to learn how a statistical quantity maps to a research question, how to select the correct comparison group, and how to interpret output rather than merely generate it.

**Assessment note.** There will be four assignments that are worth 10% each, the midterm exam is worth 20%, the final exam is worth 30%, and participation is worth 10%.

# 1 | Lecture 2: Random Variables

*Distributions • Expectation • Variance • Standard deviation • The normal distribution*

## 1.1 Executive summary

This lecture moves from probability over events to probability over numerical quantities. A random variable assigns a number to every outcome in a sample space; once that mapping is fixed, a probability distribution describes how mass or density is allocated across its possible values. Expected value locates the distribution's center; variance and standard deviation quantify its spread; and, when a normal model is appropriate, standardization converts any such distribution onto a single universal reference scale.

A handful of running examples carry the ideas. MBA salaries and automobile fuel efficiency motivate empirical distributions, binning, and histograms. A ranking problem shows how the chain rule generates a discrete probability mass function from first principles. A three-dice casino game demonstrates that a seemingly balanced payoff can still carry a negative expected return, because a small class of outcomes — triples — quietly transfers probability to the house. Two idealized lotteries with equal means but very different spreads motivate why variance and covariance matter, and simple uniform and triangular densities make the mechanics of continuous probability concrete before the normal distribution is introduced. Throughout, generative AI supplies a running motivation: a language model's next token is a random variable, softmax is its probability mass function, and decoding is sampling.

**Core principle.** A mean alone is not a complete description. Two populations or decisions can have the same expected value but very different variability. Statistical comparisons therefore require both a measure of center and a measure of uncertainty or spread.

### 1.1.1 Learning objectives

1. Define a random variable as a function from outcomes to real numbers, and distinguish discrete from continuous random variables.
2. Construct probability mass functions, probability density functions, and cumulative distribution functions, and distinguish density from probability.
3. Build empirical probability masses from counts and explain how bin choices affect a histogram.
4. Compute expected values as probability-weighted averages, including for transformations of a random variable.

5. Use the probability chain rule to derive a discrete distribution from first principles.
6. Calculate the expected return and house edge of a finite game, and the expected value and variance of simple lotteries.
7. Compute variance and standard deviation, interpret their units, and apply the transformation rules for affine functions of a random variable.
8. Distinguish independence from zero covariance, and compute the variance of a sum of dependent random variables.
9. Standardize a normal random variable and compute probabilities using the standard normal table.
10. Distinguish population parameters from the sample estimators used in later statistical inference.
11. Explain the empirical-distribution (plug-in) principle behind non-parametric estimation, and why it requires a genuinely random sample.

## 1.2 Random Variables as Numerical Mappings

### 1.2.1 Operational and formal definitions

An event is a subset of a sample space; a random variable goes one step further and assigns a numerical value to every elementary outcome. Formally, if $\Omega$ is the sample space, a random variable $X$ is a function

$$X : \Omega \to \mathbb{R}, \qquad \omega \mapsto X(\omega). \tag{1.1}$$

The word *random* describes the outcome $\omega$ selected from $\Omega$, not the function itself: once $\omega$ is known, $X(\omega)$ is determined. For a coin toss, one convenient coding is $X(H) = 1$ and $X(T) = 0$; for height, an outcome is a person and $X(\omega)$ is that person's height in centimeters; for gender or another category, numeric codes are simply labels, and assigning 0 and 1 does not by itself create an ordering, a distance, or a causal interpretation. A chosen encoding must be applied consistently once selected, even though another valid encoding of the same experiment could exist.

**Technical extension.** In a fully formal probability model $(\Omega, \mathcal{F}, P)$, $X$ must be *measurable*: for every Borel set $B$ on the real line, the preimage $\{\omega : X(\omega) \in B\}$ belongs to $\mathcal{F}$. This condition is what makes statements such as $P(X \leq x)$ well-defined probabilities.

### 1.2.2 From outcomes to a random variable

The sample space $\Omega$ and the events built from it live in the world of the experiment itself — coin faces, dice outcomes, tokens. To do calculus and statistics with these outcomes, we need numbers. A *random variable* $X$ is a function $X : \Omega \to \mathbb{R}$ that assigns a real number $X(\omega)$ to every elementary outcome $\omega$; it does not change the underlying probability law on $\Omega$, it only relabels outcomes as numbers. Crucially, the probability of a numerical statement about $X$ is defined by pulling the statement back to $\Omega$ through the *inverse* *pre-image* of $X$: for a real number $x$,

$$F_X(x) = P(X \le x) = P\Big(X^{-1}((-\infty, x])\Big) = P(\{\omega \in \Omega : X(\omega) \le x\}),$$

so $F_X$, the *cumulative distribution function* of $X$, is computed entirely from the probability law already defined on events of $\Omega$; nothing new is added except the change of coordinates from outcomes to numbers. In the die example, $X(\omega) = \omega$ recovers the familiar $1, \ldots, 6$ labeling; in the LLM example, $X$ could map a token to its rank or embedding coordinate, while $F_X$ still traces back, via $X^{-1}$, to probabilities the softmax already assigned over $\mathcal{V}$.

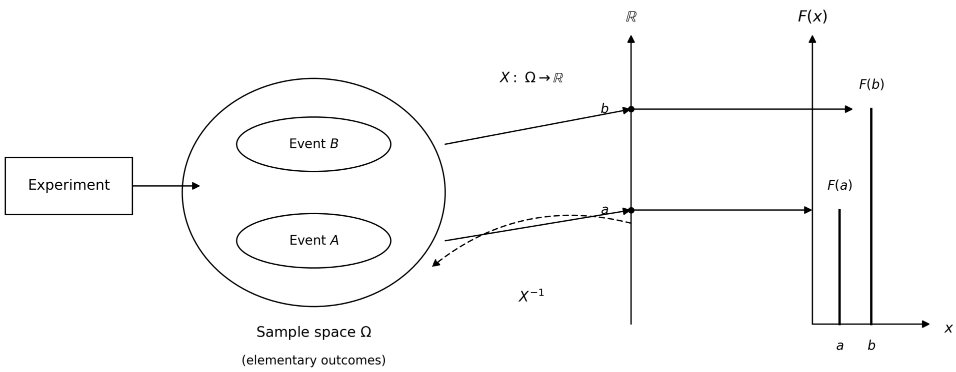


Figure 1.1: A random variable $X$ maps elementary outcomes in $\Omega$ to real numbers; the distribution function $F_X(x)$ is defined by pulling back through $X^{-1}$ to the probability law already defined on $\Omega$.

### 1.2.3 Discrete and continuous variables

Table 1.1 summarizes the two principal classes of random variable.

Table 1.1: Two principal classes of random variables

| **Class** | **Possible values** | **Examples** | **Probability representation** |
|---|---|---|---|
| Discrete | Finite or countably infinite | Coin indicator, coded category, rank, casino payoff, number of children, a language token's integer identifier | Probability mass function $p_X(x)$ |
| Continuous | An interval or other uncountable set | Height, salary as measured, fuel efficiency, time, temperature, a financial return, a normalized pixel intensity | Density $f_X(x)$ and distribution function $F_X(x)$ |

Measured continuous variables are often stored with finite precision, but the conceptual model may still be continuous; conversely, counts such as the number of purchases are discrete even when they take many values. The mathematical representation should follow the data-generating quantity, not merely the spreadsheet format.

### 1.2.4 A generative-AI motivating example

Text-generating models convert text into tokens, associate each token with a discrete numerical identifier, and generate a conditional probability distribution over the next token — a genuinely discrete random variable, however large its support. Image, audio, and video systems, by contrast, often represent data as high-dimensional numerical arrays; although stored pixels are quantized on a computer, mathematical models frequently treat latent variables or normalized intensities as continuous. For an RGB image with height $H$ and width $W$, a simple representation contains three values per pixel, so

$$\begin{aligned} \text{dimension of an } H \times W \text{ RGB image} &= 3HW; \\ \text{for } H = W = 256,\ \text{dimension} &= 196{,}608. \end{aligned} \tag{1.2}$$

The resulting joint random vector is enormously higher-dimensional than a single coin toss, which helps explain why training and inference on such models are computationally expensive.

## 1.3 Probability Mass, Density, and Cumulative Probability

### 1.3.1 Definitions

Every probability model must be nonnegative and normalized. For a discrete random variable, the *probability mass function* (PMF) assigns a probability to each attainable value, and the masses must sum to one:

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in S_X} p_X(x) = 1. \tag{1.3}$$

For a continuous random variable, the *probability density function* (PDF) $f_X$ is also nonnegative, but normalization is expressed as an integral rather than a sum, and probability is obtained by integrating the density over an interval:

$$\begin{aligned} f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1, \\ P(a \leq X \leq b) = \int_a^b f_X(x)\,dx. \end{aligned} \tag{1.4}$$

A central technical distinction is that *density is not probability*: a density may exceed one when its support is narrow, whereas probabilities are areas under the density curve and must lie between zero and one. For a genuinely continuous variable, the probability of any exact point is zero, even when the density at that point is positive: $P(X = x_0) = 0$.

The *cumulative distribution function* (CDF) works for either discrete or continuous variables and answers a directly interpretable question — what is the probability that $X$ does not exceed $x$:

$$F_X(x) = P(X \leq x). \tag{1.5}$$

A related terminological distinction is worth making precise. The word *likelihood* is sometimes used loosely for the height of a density, but density and likelihood answer different questions. A density $f(x \mid \theta)$ is a function of possible data $x$ when the parameter $\theta$ is held fixed; a likelihood $L(\theta \mid x)$ treats the same mathematical expression as a function of the unknown parameter $\theta$, with the data held fixed. Density describes curve height as data vary; likelihood is reserved for parameter inference.

### 1.3.2 The uniform distribution

The simplest continuous example is the uniform distribution. If $X$ is uniform on the interval from $a$ to $b$, the density is constant inside the interval and zero outside it; because the enclosed rectangle must have area one, its height is the reciprocal of the interval length:

$$X \sim \text{Uniform}(a,b) \implies f_X(x) = \frac{1}{b-a} \text{ for } a \le x \le b, \quad \text{and } f_X(x) = 0 \text{ otherwise.} \tag{1.6}$$

For Uniform$(1,10)$, the support has length nine, so the density is $1/9$; a proposed constant height of five would be positive but invalid, since its area would be $5 \times 9 = 45 \neq 1$. The probability of landing in the narrow interval from 7 to 7.1 is the area of a thin rectangle:

$$P(7 \le X \le 7.1) = \frac{7.1-7}{10-1} = \frac{0.1}{9} = \frac{1}{90} \approx 0.0111.$$

A narrower uniform distribution can have density greater than one without any contradiction. Uniform$(1,1.1)$ has width 0.1 and density 10; this exceeds one but is still valid because $10 \times 0.1 = 1$, and, for instance, $P(1.05 \le X \le 1.06) = 10(0.01) = 0.10$.

### 1.3.3 Empirical distributions and histograms

With observed values $x_1, \dots, x_N$, divide the measurement axis into bins $B_j = [b_j, b_{j+1})$. The count $n_j$ records how many observations fall in bin $j$, and dividing by $N$ converts counts into an empirical probability mass:

$$n_j = \sum_{i=1}^{N} \mathbf{1}\{x_i \in B_j\}, \qquad \hat{p}_j = \frac{n_j}{N}. \tag{1.7}$$

For equal-width bins, bar heights may display counts or proportions directly. For unequal widths, a proper density histogram should use height $n_j/(Nw_j)$, where $w_j$ is the bin width, so that bar *area* — not height alone — equals probability. A histogram is therefore an estimate or grouped representation of a distribution, not literally the underlying probability density function.

Two applied datasets illustrate the construction: an MBA salary dataset of 2,597 anonymized, inflation-adjusted records, and an automobile dataset of 392 vehicle models produced from roughly 1970 to 1982. Table 1.2 contrasts the distinct questions that different summaries of such a distribution answer, and Table 1.3 previews what a standard descriptive-statistics call reports before the two datasets are worked through in code below.

Table 1.2: Distinct questions answered by common summaries

| Summary | Question answered | Sensitivity |
|---|---|---|
| Exact mode | Which recorded value occurs most often? | Sensitive to rounding and repeated values |
| Modal bin | Which interval contains the most observations? | Sensitive to bin origin and width |
| Median | Where is the 50th percentile? | Robust to extreme values |
| Mean | Where is the probability-weighted balance point? | Pulled toward extreme (e.g. very high salary) values |

Table 1.3: Typical descriptive-statistics output

| Output | Mathematical meaning | Interpretive use |
|---|---|---|
| count | Number of nonmissing observations | Data completeness |
| mean | Arithmetic sample average | Estimated center |
| std | Sample standard deviation | Estimated spread |
| min / max | Observed extremes | Range and possible anomalies |
| 25% / 50% / 75% | Sample quartiles | Robust location and spread |

**1.3.3.0.1 The MBA salary dataset in Python.** Loading the file and calling `describe()` gives the summary in Table 1.4:

```
import numpy as np
import pandas as pd

salary = pd.read_excel("M2 Salary Data.xlsx", sheet_name="DataTable")
salary.describe()
```

Table 1.4: `salary.describe()` output

| Statistic | Salary ($) |
|---|---:|
| count | 2,597 |
| mean | 111,208.37 |
| std | 47,515.30 |
| min | 33,000.00 |
| 25% | 73,930.80 |
| 50% (median) | 99,684.00 |
| 75% | 140,000.00 |

Table 1.4: `salary.describe()` output

| **Statistic** | **Salary ($)** |
|---|---|
| max | 336,966.00 |

The distribution is right-skewed: the mean, $111,208, exceeds the median, $99,684, by more than $11,000 — exactly the asymmetry a right-skewed monetary variable produces, since every value enters the mean in proportion to its magnitude while the median only counts rank. A symmetric normal model should never be assumed merely because a histogram happens to have one central peak.

Binning the salary variable into $20,000 intervals and normalizing the counts gives the empirical PMF in Table 1.5:

```
bins = range(30000, 350000, 20000)
salary["Bins"] = pd.cut(salary["Salary"], bins)
salary["Bins"].value_counts(normalize=True).sort_index()
```

Table 1.5: Empirical PMF of binned MBA salary

| **Bin** | $\widehat{p}_j$ |
|---|---|
| (30,000, 50,000] | 2.7% |
| (50,000, 70,000] | 18.3% |
| (70,000, 90,000] | 20.8% |
| (90,000, 110,000] | 16.4% |
| (110,000, 130,000] | 12.0% |
| (130,000, 150,000] | 9.0% |
| (150,000, 170,000] | 6.6% |
| (170,000, 190,000] | 7.3% |
| (190,000, 210,000] | 4.1% |
| (210,000, 230,000] | 1.4% |
| (230,000, 250,000] | 0.5% |
| (250,000, 270,000] | 0.4% |
| (270,000, 290,000] | 0.2% |
| (290,000, 310,000] | 0.2% |
| (310,000, 330,000] | 0.1% |
| (330,000, 350,000] | 0.0% |

The most populated interval, $70,000–$90,000, holds about 20.8% of the sample — the "about one-fifth" figure quoted informally in lecture. Note also that `pandas`' default binning convention, used above, constructs *right-closed* intervals $(b_j, b_{j+1}]$, the opposite of the left-closed convention $[b_j, b_{j+1})$ used in Equation (1.7); either convention is valid so long as it is documented, since a value sitting exactly on a boundary is assigned differently under each rule. Because the `describe()` output above shows the salary range running from $33,000 to $336,966, entirely inside the chosen bin range of $30,000 to $350,000, every observation lands in some bin here and none are silently dropped.

**1.3.3.0.2 The automobile fuel-efficiency dataset in Python.** The same workflow bins miles per gallon (MPG) into intervals of width five:

```
auto = pd.read_excel("M2 Auto_data.xlsx", sheet_name="Original Dataset")
bins = range(0, 50, 5)
auto["bins"] = pd.cut(auto["mpg"], bins)
auto["bins"].value_counts(normalize=True).sort_index()
```

Table 1.6: Empirical PMF of binned automobile fuel efficiency (MPG)

| **Bin** | $\widehat{p}_j$ |
|---|---|
| (0, 5] | 0.0% |
| (5, 10] | 0.8% |
| (10, 15] | 16.9% |
| (15, 20] | 23.3% |
| (20, 25] | 19.4% |
| (25, 30] | 18.7% |
| (30, 35] | 12.8% |
| (35, 40] | 6.4% |
| (40, 45] | 1.8% |

The distribution is unimodal, centered around 15–25 MPG, and the proportions sum to 100% up to rounding, confirming that every recorded MPG value between 0 and 45 was captured by the chosen bins. Section 1.6 returns to this table to compute an approximate grouped-data mean directly from these bin proportions.

**Choosing bins responsibly.** Both examples above avoided trouble only because their bin edges happened to cover the full observed range — a fact confirmed here by checking, not assumed. In general: always check the minimum and maximum against the chosen edges (Python's `range` and `arange` both exclude the stop value, a common source of a silently truncated top bin); keep widths equal when comparing bar heights directly as probability masses; test whether a conclusion changes under reasonable alternative widths or starting points; report the sample size and bin definitions so the figure can be reproduced; and never treat a smooth, bell-like appearance as proof of normality.

## 1.4 Sampling, Empirical Frequency, and Reproducibility

A *sample* is a realized value from a random variable; a dataset of size $n$ contains $n$ such draws, which can be generated in Excel or Python from a specified distribution. If the draws are independent and identically distributed, the empirical proportion of observations falling in an event $A$ estimates its model probability:

$$\widehat{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \in A\}. \tag{1.8}$$

In one set of 100 simulated draws from Uniform$(1, 1.1)$, seven fell between 1.05 and 1.06, giving an empirical proportion of 0.07 against the theoretical value 0.10 found above.

This is not a contradiction: sampling error is expected in a finite dataset, and under the **law of large numbers** the empirical proportion converges to the true probability as $n$ grows,

$$\widehat{P}_n(A) \to P(A) \quad \text{as } n \to \infty,$$

under standard independent-sampling assumptions. A *pseudorandom seed* initializes a deterministic number generator; reusing the same seed with the same generator and software implementation reproduces the same sequence, which is valuable for validation and debugging, though cross-platform results may differ when implementations differ. A seed does not make observations more random — it makes a pseudorandom experiment reproducible.

This machinery is closely connected to AI inference. A generative model produces outputs by sampling from learned conditional distributions, though the analogy should not be taken too literally: a modern language model does not merely transform one scalar uniform distribution into English. It learns a high-dimensional parameterized network, computes conditional token probabilities with a temperature-scaled softmax layer, and applies a decoding rule such as greedy selection, temperature sampling, top-$k$ sampling, or nucleus sampling:

$$P(\text{next token} = j \mid \text{context}) = \frac{\exp(z_j/T)}{\sum_k \exp(z_k/T)}. \tag{1.9}$$

Here $z_j$ is the model's score for token $j$ and $T$ is the temperature. Lower temperature concentrates probability on high-scoring tokens; higher temperature flattens the distribution and increases output diversity. This is the operational link between probability distributions, sampling, and why the same prompt can produce different responses on different runs.

### 1.4.1 The empirical distribution and non-parametric (plug-in) estimation

The estimator $\widehat{P}_n(A)$ in Equation (1.8) is a special case of a much more general idea, the **plug-in** or **non-parametric** estimation principle: whenever a population quantity can be written as an expectation under the population distribution $P(x)$, and $P(x)$ itself is unknown, replace $P(x)$ by the *empirical distribution* that assigns weight $1/n$ to each of the $n$ sampled observations. No parametric family — normal, uniform, or otherwise — needs to be assumed for this to work; the estimator is built directly from the data.

Concretely, the population expectation $E[X] = \sum_x x\, P(x)$ becomes, after substituting the empirical weights $1/n$ for $P(x)$, exactly the sample mean:

$$\overline{X} = \sum_{i=1}^{n} x_i \cdot \frac{1}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i. \tag{1.10}$$

The same substitution applied to a variance, a proportion, or any other functional of the distribution produces the corresponding sample statistic. This is why ordinary sample means and sample variances are special cases of the broader **Monte Carlo principle**: random sampling lets us approximate an expectation under $P(x)$ without ever writing $P(x)$ down explicitly.

For an MBA salary population of more than 2,000 observations with population mean about \$111,208, a random sample of $n = 100$ produced a sample mean of about \$111,568. The two numbers are not identical, and should not be: the empirical distribution built from any one finite sample only approximates the population distribution, and a different sample of the same size would give a different sample mean. The law of large numbers guarantees convergence of this approximation as $n$ grows, not exact agreement at any fixed $n$.

**Why random sampling matters.** The plug-in principle is only as good as the sample it is built on. If the sampling mechanism systematically favors part of the population — surveying only certain neighborhoods, or observations that happen to be easiest to collect — the empirical distribution can converge to the wrong target no matter how large $n$ becomes: a large biased sample can yield a very precise estimate of the wrong quantity. Before applying any estimator in this chapter, it is worth asking how the data were actually collected: were observations selected independently, and does observing one unit change the chance that another is observed?

## 1.5 Transformations of Random Variables

### 1.5.1 Affine transformations

A transformation constructs a new random variable from an existing one: if $Y = g(X)$, each realized $X$ value is converted by the deterministic function $g$. The most important basic case is an **affine transformation**,

$$Y = aX + b.$$

The coefficient $a$ changes scale — stretching the distribution if $|a| > 1$, compressing it if $0 < |a| < 1$, and reversing its orientation if $a < 0$ — while the constant $b$ shifts every value by the same amount. If $X$ has support from a lower bound $L$ to an upper bound $U$ and $a > 0$, then $Y \in [aL + b,\ aU + b]$.

As a worked example, map Uniform$(1, 10)$ onto Uniform$(4, 6)$ in three steps: subtract one (moving the support from 0 to 9), multiply by $2/9$ (compressing its length from 9 to 2), and add four (shifting the support to 4 through 6):

$$Y = 4 + \frac{2}{9}(X - 1) = \frac{2}{9}X + \frac{34}{9}.$$

Checking endpoints verifies the mapping: $X = 1 \Rightarrow Y = 4$, and $X = 10 \Rightarrow Y = 6$. Because the transformation is linear and increasing and the input is uniform, the output is also uniform, with density $f_Y(y) = 1/2$ for $4 \le y \le 6$.

For a differentiable, one-to-one transformation $Y = g(X)$, density changes according to the Jacobian rule, which preserves total probability while horizontal distances are stretched or compressed:

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy}g^{-1}(y)\right|. \tag{1.11}$$

A valid transformation need not be one-to-one; $Y = X^2$, for instance, is perfectly valid. One-to-one behavior is required only for the simplest single-inverse change-of-variables formula above. If several $X$ values map to the same $Y$, their density contributions must be summed over all inverse branches:

$$f_Y(y) = \sum_{x:\, g(x)=y} \frac{f_X(x)}{|g'(x)|}, \qquad \text{when the inverse branches are regular.}$$

### 1.5.2 The Jacobian rule via an affine transformation

Suppose $X$ has known density $f_X(x)$ and we form a new random variable through an invertible affine map $Y = g(X) = aX + b$, with $a \neq 0$. Since $g$ is monotonic, it has an inverse $g^{-1}(y) = (y - b)/a$, and the density of $Y$ is obtained by the *change-of-variables (Jacobian) rule*:

$$f_Y(y) \;=\; f_X\big(g^{-1}(y)\big) \left|\frac{d}{dy} g^{-1}(y)\right| \;=\; f_X\!\left(\frac{y-b}{a}\right) \frac{1}{|a|}.$$

The Jacobian factor $|dx/dy| = 1/|a|$ is not optional bookkeeping: it is exactly what keeps probability mass conserved when the transformation stretches or compresses the sample space. A narrow interval $[x, x + dx]$ under $X$ maps to a wider interval $[y, y + a\,dx]$ under $Y$ (when $|a| > 1$); since the probability of landing in that interval cannot change, the density must drop by the same factor $1/|a|$ that the interval width grew by.

Concretely, take $X \sim N(0, 1)$, so $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and let $Y = 2X + 3$. Applying the rule with $a = 2$, $b = 3$:

$$f_Y(y) = f_X\!\left(\frac{y-3}{2}\right) \frac{1}{2} = \frac{1}{2\sqrt{2\pi}} \exp\!\left(-\frac{(y-3)^2}{8}\right),$$

which is exactly the density of $N(3, 4)$ — mean shifted by $b = 3$, variance scaled by $a^2 = 4$. Figure 1.2 shows this side by side: the shaded interval under $f_X$ and its image under $f_Y$ enclose the same area, even though the interval under $Y$ is twice as wide and half as tall.

### 1.5.3 A non-affine transformation: turning a flat density into a curved one

The affine case had a constant Jacobian factor $1/|a|$ because the map stretched every interval by the same amount, everywhere. A non-affine (nonlinear) map breaks that uniformity: the local stretch factor $|g'(x)|$ now depends on *where* $x$ is, and this position-dependence is precisely what turns a flat density into a curved one.

Let $X \sim \text{Uniform}(0, 1)$, so $f_X(x) = 1$ for $x \in (0, 1)$, and define $Y = g(X) = -\ln X$. Since $g$ is strictly decreasing on $(0, 1)$, it is invertible: $g^{-1}(y) = e^{-y}$, with derivative $\frac{d}{dy} g^{-1}(y) = -e^{-y}$. The Jacobian rule gives

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \left|\frac{d}{dy} g^{-1}(y)\right| = 1 \cdot e^{-y} = e^{-y}, \qquad y > 0,$$

so $Y \sim \text{Exponential}(1)$: a nonlinear function of a uniform variable produces the exponential distribution. Equivalently, writing the rule as $f_Y(y) = f_X(x)/|g'(x)|$ with

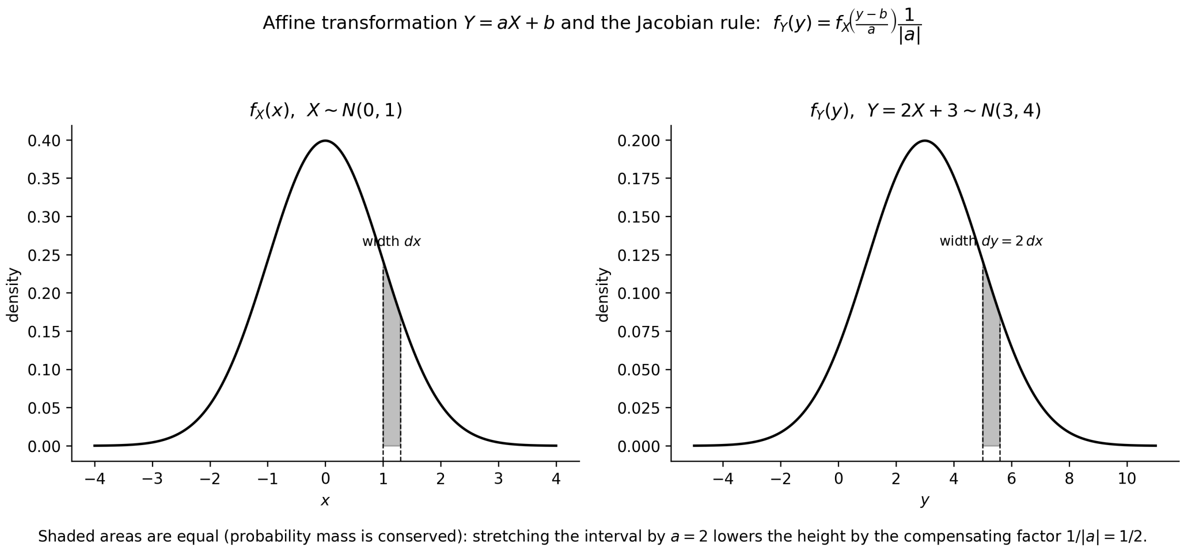


Figure 1.2: An affine transformation $Y = 2X + 3$ applied to $X \sim N(0, 1)$. Stretching the interval by $a = 2$ is exactly compensated by the Jacobian factor $1/|a| = 1/2$ in the density, so probability mass is conserved.

$g'(x) = -1/x$, the local stretch factor is $1/x$ — large near $x = 0$, close to 1 near $x = 1$. Figure 1.3 makes this visible: an interval near $x = 0.1$ (orange) is stretched by a factor of about 8 under $g$, so the corresponding piece of $f_Y$ drops sharply, while an interval near $x = 0.8$ (blue) is stretched by only about 1.2, leaving $f_Y$ nearly unchanged.

This construction is exactly *inverse transform sampling*: for any target CDF $F$, applying $Y = F^{-1}(U)$ to $U \sim \text{Uniform}(0, 1)$ produces a random variable with distribution $F$. The exponential case above is the special case $F^{-1}(u) = -\ln(1-u)$ (here written with $u \leftrightarrow 1-u$ for convenience), and the same idea underlies sampling a token from the categorical distribution a language model outputs after softmax — a nonlinear reshaping of a uniform draw into the target law.

### 1.5.4 Uniform and triangular densities

A density is valid whenever it is nonnegative and its total area is one — a criterion that extends naturally beyond rectangles. For a triangular density with base from 1 to 10 (base length nine) and peak height $h$, normalization requires

$$\tfrac{1}{2}(9)h = 1 \quad \Longrightarrow \quad h = \tfrac{2}{9}.$$

Moving the triangular peak changes where probability mass concentrates but not the height needed for normalization, as long as the base endpoints stay fixed at 1 and 10 and the triangle returns to zero at both ends. A peak near the left creates a right-skewed distribution with a long right tail; a peak near the right creates a left-skewed distribution; a peak at the midpoint 5.5 produces a symmetric triangular distribution, for which the mean and median coincide at 5.5. For a skewed triangular distribution the mode, median, and mean generally differ, since these three measures answer different questions and need not coincide: the mode is the location of the density peak, the median divides probability mass in half, and the mean is the balance point.

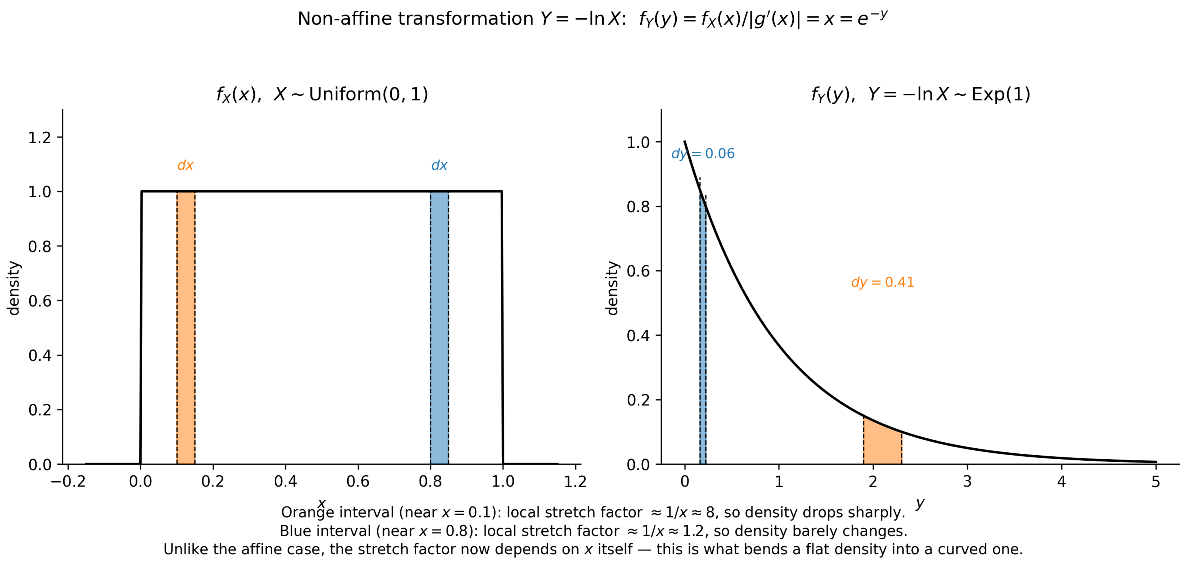


Figure 1.3: A non-affine transformation $Y = -\ln X$ of $X \sim \text{Uniform}(0, 1)$ produces $Y \sim \text{Exponential}(1)$. Unlike the affine case, the local stretch factor $1/x$ varies with $x$, bending the flat density into a curved one.

### 1.5.5 Quantiles and the discrete median convention

The median is a quantile: a continuous median $m$ satisfies equal probability on either side when the density has no flat ambiguity, and more generally a median satisfies

$$P(X \le m) \ge 0.5 \qquad \text{and} \qquad P(X \ge m) \ge 0.5.$$

Using the CDF $F_X(x) = P(X \le x)$, the course convention for a *discrete* median is the smallest attainable value whose cumulative probability reaches or exceeds one half:

$$m = \inf\{x : F_X(x) \ge 0.5\}.$$

For a family-size distribution assigning probabilities 0.15, 0.20, 0.35, and 0.30 to 0, 1, 2, and 3 children, the cumulative probabilities are 0.15, 0.35, 0.70, and 1. Since $F(1) = 0.35 < 0.5$ while $F(2) = 0.70 \ge 0.5$, the median is 2 children.

## 1.6 Expected Value

### 1.6.1 Probability-weighted center

The expected value, also called the mean and written $\mu$, is the probability-weighted average of a random variable's possible outcomes — a sum for a discrete variable, an integral against the density for a continuous one:

$$\begin{aligned} E[X] &= \sum_x x \, p_X(x) \quad \text{(discrete)}, \\ E[X] &= \int_{-\infty}^{\infty} x \, f_X(x) \, dx \quad \text{(continuous)}. \end{aligned} \tag{1.12}$$

A center-of-mass interpretation is useful: imagine each possible value placed on a balance beam with mass equal to its probability, and the expected value is the point at which the beam balances.

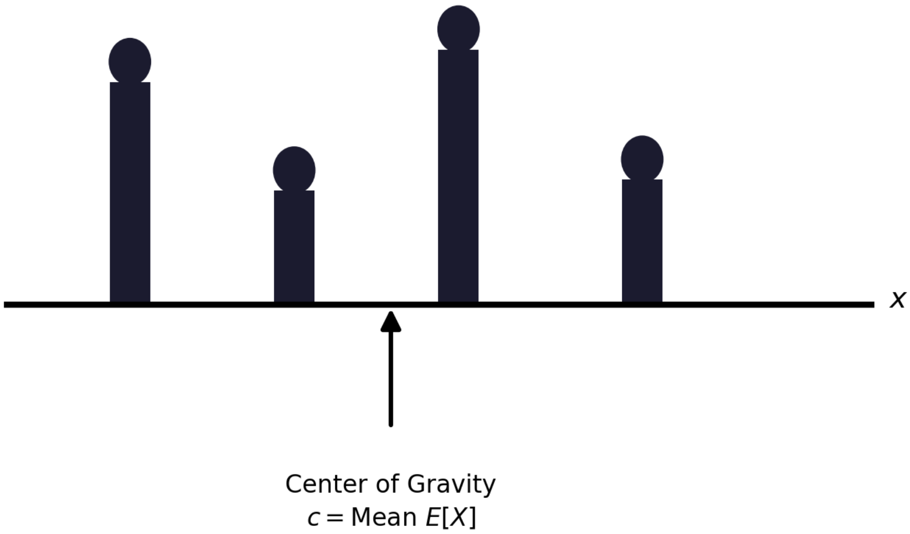


Figure 1.4: Interpretation of the mean as a center of gravity. Given a bar with a weight $p_X(x)$ placed at each point $x$ with $p_X(x) > 0$, the center of gravity $c$ is the point at which the sum of the torques from the weights to its left are equal to the sum of the torques from the weights to its right.

That is,

$$\sum_x (x - c)\, p_X(x) = 0, \qquad \text{or} \qquad c = \sum_x x\, p_X(x), \tag{1.13}$$

and the center of gravity is equal to the mean $E[X]$.

Unlike the discrete median, the expected value need not be an attainable outcome — a family cannot have 1.8 children, and no die shows 3.5, yet both are meaningful population averages. For the family-size distribution above,

$$E[X] = 0(0.15) + 1(0.20) + 2(0.35) + 3(0.30) = 1.80.$$

Many research quantities are transformations $g(X)$ rather than $X$ itself — expected utility, squared error, and profit are examples. The *law of the unconscious statistician* computes such an expectation directly from the distribution of $X$, without first deriving the distribution of $g(X)$:

$$E[g(X)] = \sum_x g(x)\, p_X(x). \tag{1.14}$$

Expectation is linear even when variables are dependent, and constants move through the expectation operator in the natural way:

$$E[aX + b] = aE[X] + b, \qquad E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n]. \tag{1.15}$$

For 1,000 families with the family-size distribution above, the expected total number of children is $1000(1.8) = 1800$; if each child is independently a girl with probability one half, the expected number of girls is 900. Linearity alone supplies the expected total — independence is not required for the addition of expectations, although the one-half conditional assumption is needed for the gender calculation.

We finally illustrate by example a common pitfall: unless $g(X)$ is a linear function, it is not generally true that $E[g(X)]$ is equal to $g\big(E[X]\big)$.

**Example. Average Speed Versus Average Time.** If the weather is good (which happens with probability 0.6), Alice walks the 2 miles to class at a speed of $V = 5$ miles per hour, and otherwise drives her motorcycle at a speed of $V = 30$ miles per hour. What is the mean of the time $T$ to get to class?

The correct way to solve the problem is to first derive the PMF of $T$,

$$p_T(t) = \begin{cases} 0.6 & \text{if } t = 2/5 \text{ hours,} \\ 0.4 & \text{if } t = 2/30 \text{ hours,} \end{cases}$$

and then calculate its mean by

$$E[T] = 0.6 \cdot \frac{2}{5} + 0.4 \cdot \frac{2}{30} = \frac{4}{15} \text{ hours.}$$

However, it is wrong to calculate the mean of the speed $V$,

$$E[V] = 0.6 \cdot 5 + 0.4 \cdot 30 = 15 \text{ miles per hour,}$$

and then claim that the mean of the time $T$ is

$$\frac{2}{E[V]} = \frac{2}{15} \text{ hours.}$$

To summarize, in this example we have

$$T = \frac{2}{V}, \qquad \text{and} \qquad E[T] = E\left[\frac{2}{V}\right] \neq \frac{2}{E[V]}.$$

When only histogram bins are available, replace each observation in bin $j$ by a representative value $m_j$, usually the midpoint, giving an approximate grouped-data mean:

$$\widehat{\mu} \approx \sum_j m_j \, \widehat{p}_j. \tag{1.16}$$

**Practical caution.** Using lower bin boundaries instead of midpoints systematically shifts this approximation downward. Whenever the original observations are available, calculate the mean from the raw values and reserve the histogram for visualization rather than estimation.

Applying Equation (1.16) to the automobile MPG bins of Table 1.6, using each bin's midpoint $m_j$ (e.g. $m = 2.5$ for $(0, 5]$, $m = 7.5$ for $(5, 10]$, and so on up to $m = 42.5$ for $(40, 45]$) as the representative value:

$$\begin{aligned} \widehat{\mu} \approx\ & 2.5(0.000) + 7.5(0.008) + 12.5(0.169) + 17.5(0.233) + 22.5(0.194) \\ & + 27.5(0.187) + 32.5(0.128) + 37.5(0.064) + 42.5(0.018) \approx 23.1 \text{ MPG.} \end{aligned}$$

This grouped-data estimate is only approximate — it replaces every observation inside a bin with that bin's midpoint — but it requires nothing beyond the histogram itself, which is exactly the setting in which Equation (1.16) is used in practice.

### 1.6.2 Law of Iterated Expectations

If $Y$ is a random variable that partitions the sample space into cases, the mean of $X$ can be computed by first averaging $X$ within each case, and then averaging those conditional means over the distribution of $Y$:

$$E[X] = \sum_y E[X \mid Y = y]\, P(Y = y),$$

or more compactly, $E[X] = E\big[E[X \mid Y]\big]$. The inner quantity $E[X \mid Y]$ is itself a random variable — a function of $Y$ — and the outer expectation averages it over $Y$'s own distribution. This is also called the *law of total expectation* or the *tower property*, and it is often the easiest way to compute a mean when conditioning on some event or random variable makes the calculation simple in each case.

**Example. Highest Rank of a Woman, via Iterated Expectation.** Two men and two women are ranked according to their scores on an exam. Assume no two scores are alike, so all 4! possible rankings are equally likely, and let $X$ denote the highest (numerically smallest) ranking achieved by a woman. We compute $E[X]$ by conditioning on the gender of the top-ranked (1st place) person.

**Conditional Expectation Solution.** Conditional expectation gives a very clean solution.

Let $X$ be the rank of the highest-ranked woman. Since there are 2 men and 2 women,

$$X \in \{1, 2, 3\}.$$

We want $E[X]$. Instead of immediately calculating the full distribution of $X$, condition on the gender of the person ranked first.

Let

$$A = \{\text{the first-ranked person is a woman}\}.$$

Then by the law of total expectation,

$$E[X] = E[X \mid A]P(A) + E[X \mid A^c]P(A^c).$$

Because there are 2 women and 2 men,

$$P(A) = \frac{2}{4} = \frac{1}{2}, \qquad P(A^c) = \frac{1}{2}.$$

**Case 1: The first-ranked person is a woman.** If the first-ranked person is a woman, then automatically the highest-ranked woman has rank 1. Therefore,

$$E[X \mid A] = 1.$$

**Case 2: The first-ranked person is a man.** Now suppose the first-ranked person is a man. After removing that man, there are 3 people left:

$$W, \quad W, \quad M.$$

We need the first woman among these remaining three people.

At rank 2, the person is a woman with probability

$$\frac{2}{3}.$$

If that happens,

$$X = 2.$$

The person at rank 2 is a man with probability

$$\frac{1}{3}.$$

But since there were only two men total, if the first two positions are both occupied by men, then the next person must be a woman. Therefore,

$$X = 3.$$

Hence,

$$E[X \mid A^c] = 2\left(\frac{2}{3}\right) + 3\left(\frac{1}{3}\right).$$

So,

$$E[X \mid A^c] = \frac{4}{3} + 1 = \frac{7}{3}.$$

Now apply the law of total expectation:

$$E[X] = 1\left(\frac{1}{2}\right) + \frac{7}{3}\left(\frac{1}{2}\right).$$

Therefore,

$$E[X] = \frac{1}{2} + \frac{7}{6} = \frac{3}{6} + \frac{7}{6} = \boxed{\frac{5}{3}}.$$

Thus the expected rank of the highest-ranked woman is

$$\boxed{E[X] = \frac{5}{3} \approx 1.667}.$$

This is a useful example of conditional expectation because we break the problem into two much simpler situations according to whether the top-ranked person is a woman or a man.

**Generic version: Repeated Conditional Expectation.** This is a particularly elegant way to solve the problem using repeated conditional expectation. The key is to write the expected rank as

$$\boxed{E[X] = 1 + \text{expected additional rank after the first person}}$$

and repeat the same argument after observing a man.

Let $E_{m,w}$ denote the expected rank of the first woman when there are $m$ men and $w$ women remaining. Initially,

$$E[X] = E_{2,2}.$$

**First iteration.** We always inspect at least one person, so start with 1.

With probability

$$\frac{2}{4} = \frac{1}{2},$$

the first person is a woman. We stop, so there is no additional rank.

With probability

$$\frac{2}{4} = \frac{1}{2},$$

the first person is a man. Then we have 1 man and 2 women remaining, and the expected additional number of positions is $E_{1,2}$.

Therefore,

$$\boxed{E_{2,2} = \frac{2}{4}(1) + \frac{2}{4}(1 + E_{1,2})}$$

or

$$E_{2,2} = 1 + \frac{2}{4}E_{1,2}$$

or

$$E_{2,2} = 1 + \frac{1}{2}E_{1,2}.$$

Notice how the "+1" accounts for the person we have just inspected.

**Second iteration.** Now calculate $E_{1,2}$. There are 3 people:

$$1 \text{ man}, \qquad 2 \text{ women}.$$

Again, we certainly inspect one person, giving the 1.

If that person is a woman, which happens with probability 2/3, we stop.

If that person is the remaining man, which happens with probability 1/3, we continue.

So

$$\boxed{E_{1,2} = 1 + \frac{1}{3}E_{0,2}}$$

Now there are zero men and two women. Therefore the next person must be a woman:

$$E_{0,2} = 1.$$

Hence,

$$E_{1,2} = 1 + \frac{1}{3}(1) = \frac{4}{3}.$$

**Substitute back into the first iteration.** We had

$$E_{2,2} = 1 + \frac{1}{2}E_{1,2}.$$

Therefore,

$$E_{2,2} = 1 + \frac{1}{2}\left(\frac{4}{3}\right)$$

and hence

$$\boxed{E[X] = 1 + \frac{2}{3} = \frac{5}{3}.}$$

So the expected highest rank achieved by a woman is

$$\boxed{\frac{5}{3} \approx 1.667.}$$

The entire calculation can be written very compactly as

$$\boxed{E[X] = 1 + \frac{1}{2}\left(1 + \frac{1}{3}(1)\right) = \frac{5}{3}.}$$

This recursive form is useful: in general,

$$\boxed{E_{m,w} = 1 + \frac{m}{m+w} E_{m-1,w}}$$

with boundary condition

$$E_{0,w} = 1.$$

The intuition is: count the current position as 1; only if the current person is a man do we need to add the conditional expected waiting time for the remaining people.

**Deriving the Closed-Form Formula from the Recursion.** A good way to construct the closed-form formula is to start from the recursion and expand it a few times until a pattern appears.

We have

$$E_{m,w} = 1 + \frac{m}{m+w} E_{m-1,w},$$

with

$$E_{0,w} = 1.$$

Now expand recursively:

$$E_{m,w} = 1 + \frac{m}{m+w}\left(1 + \frac{m-1}{m+w-1} E_{m-2,w}\right).$$

So

$$E_{m,w} = 1 + \frac{m}{m+w} + \frac{m(m-1)}{(m+w)(m+w-1)} E_{m-2,w}.$$

Expanding again,

$$E_{m,w} = 1 + \frac{m}{m+w} + \frac{m(m-1)}{(m+w)(m+w-1)} + \cdots$$

Eventually,

$$E_{m,w} = 1 + \frac{m}{m+w} + \frac{m(m-1)}{(m+w)(m+w-1)} + \cdots + \frac{m!}{(m+w)(m+w-1)\cdots(w+1)}.$$

This expression has a probabilistic interpretation. The second term is the probability that the first person is a man:

$$P(X > 1) = \frac{m}{m+w}.$$

The third term is the probability that the first two people are both men:

$$P(X > 2) = \frac{m}{m+w} \cdot \frac{m-1}{m+w-1}.$$

In general,

$$P(X > k) = \frac{m(m-1)\cdots(m-k+1)}{(m+w)(m+w-1)\cdots(m+w-k+1)}.$$

Therefore the recursion is really producing the tail-sum formula

$$E[X] = 1 + P(X > 1) + P(X > 2) + \cdots .$$

**Guessing the simple formula.** Compute a few small cases.

For $m = 0$,

$$E_{0,w} = 1.$$

For $m = 1$,

$$E_{1,w} = 1 + \frac{1}{w+1} = \frac{w+2}{w+1}.$$

For $m = 2$,

$$E_{2,w} = 1 + \frac{2}{w+2} \cdot \frac{w+2}{w+1} = 1 + \frac{2}{w+1} = \frac{w+3}{w+1}.$$

For $m = 3$,

$$E_{3,w} = 1 + \frac{3}{w+3} \cdot \frac{w+3}{w+1} = 1 + \frac{3}{w+1} = \frac{w+4}{w+1}.$$

So the pattern becomes clear:

$$E_{m,w} = 1 + \frac{m}{w+1}.$$

Hence,

$$\boxed{E_{m,w} = \frac{m+w+1}{w+1}}.$$

**A more intuitive construction: the gaps argument.** There is also an even more intuitive way to construct it. Imagine placing the $w$ women first. They create $w+1$ gaps for the $m$ men:

$$\boxed{\_}\ W\ \boxed{\_}\ W\ \boxed{\_}\ \cdots\ W\ \boxed{\_}$$

By symmetry, the $m$ men are distributed equally in expectation among these $w+1$ gaps. So the expected number of men before the first woman is

$$\frac{m}{w+1}.$$

The first woman's rank is one plus the number of men before her. Therefore,

$$E[X] = 1 + \frac{m}{w+1}.$$

So the closed form can be discovered either algebraically, by expanding the recursion and spotting the pattern, or probabilistically, by the $w+1$ gaps argument. The second argument explains particularly well why the denominator is $w+1$, rather than $w$.

**The Exponential Distribution: A Continuous Analogue.** The exponential distribution is a perfect example because its memoryless property gives almost exactly the same recursive structure.

Suppose
$$X \sim \mathrm{Exp}(\lambda).$$
We want to derive
$$E[X] = \frac{1}{\lambda}$$
using the same idea: spend a little bit of time, then condition on whether the event has happened.

Take a small time interval $h > 0$. During the first $h$ units of time, two things can happen. If the exponential event occurs before time $h$, we stop. If it does not occur by time $h$, then we have already waited $h$, and by memorylessness, the remaining waiting time has the same distribution as the original $X$.

So, approximately,
$$E[X] = h + P(X > h)E[X],$$
except that this is not quite exact because if the event occurs before $h$, we have waited less than $h$, not exactly $h$.

A cleaner exact version uses conditioning on survival past $h$:
$$E[X] = E[X \mid X \leq h]P(X \leq h) + E[X \mid X > h]P(X > h).$$
By memorylessness,
$$E[X \mid X > h] = h + E[X].$$
Also,
$$P(X > h) = e^{-\lambda h}.$$
Therefore,
$$E[X] = E[X \mid X \leq h](1 - e^{-\lambda h}) + (h + E[X])e^{-\lambda h}.$$
Now let
$$\mu = E[X].$$
Then
$$\mu = E[X \mid X \leq h](1 - e^{-\lambda h}) + he^{-\lambda h} + \mu e^{-\lambda h}.$$
Hence
$$\mu(1 - e^{-\lambda h}) = E[X \mid X \leq h](1 - e^{-\lambda h}) + he^{-\lambda h}.$$
So
$$\mu = E[X \mid X \leq h] + \frac{he^{-\lambda h}}{1 - e^{-\lambda h}}.$$
Now let $h \to 0$. Because $0 \leq X \leq h$ conditional on $X \leq h$,
$$E[X \mid X \leq h] \to 0.$$
Also,
$$\frac{he^{-\lambda h}}{1 - e^{-\lambda h}} \to \frac{1}{\lambda},$$

because for small $h$,

$$1 - e^{-\lambda h} \approx \lambda h.$$

Therefore,

$$\boxed{E[X] = \frac{1}{\lambda}}.$$

**A closer analogue via discretized time.** There is an even closer analogue to the earlier ranking recursion if we discretize time into tiny intervals. Let $\mu = E[X]$. Over a tiny interval $h$,

$$P(\text{survive } h) = e^{-\lambda h}.$$

Then heuristically,

$$\mu \approx h + e^{-\lambda h}\mu.$$

So

$$\mu(1 - e^{-\lambda h}) \approx h,$$

and therefore

$$\mu \approx \frac{h}{1 - e^{-\lambda h}}.$$

Taking $h \to 0$,

$$\boxed{\mu = \lim_{h \to 0} \frac{h}{1 - e^{-\lambda h}} = \frac{1}{\lambda}}.$$

This is very similar to the ranking recursion

$$E = 1 + pE_{\text{remaining}}.$$

For the exponential case, memorylessness makes the "remaining problem" identical to the original problem, which is why the recursion becomes especially simple.

**Question: Which random.variables can be calculated by using the same trick?**

**Answer: Yes**

**First-Step Analysis: A Unifying Principle.** The "same trick" is essentially a first-step recursion:

$$E[X] = \text{what happens in the first step} + E[\text{remaining problem}].$$

It works especially well when, after conditioning on the first event, the remaining problem is either identical to the original problem or belongs to the same family with updated parameters. Some important examples are:

- **Geometric distribution.** This is probably the closest discrete analogue of the exponential. If $X$ is the number of trials until the first success, with success probability $p$, then

  $$E[X] = 1 + (1-p)E[X].$$

  Therefore

  $$\boxed{E[X] = \frac{1}{p}}.$$

  This works because after a failure, you are back to exactly the same problem.

- **Negative binomial distribution.** If $E_r$ is the expected number of trials required to obtain $r$ successes, you can condition on the first trial:

  $$E_r = 1 + pE_{r-1} + (1-p)E_r.$$

  Hence

  $$E_r = \frac{1}{p} + E_{r-1},$$

  which recursively gives

  $$\boxed{E_r = \frac{r}{p}}.$$

- **Exponential distribution.** As we just discussed, memorylessness means that after surviving for some time, the remaining waiting time has the same distribution. This ultimately gives

  $$\boxed{E[X] = \frac{1}{\lambda}}.$$

- **Poisson-process waiting times.** The waiting time until the $r$-th Poisson arrival is Gamma/Erlang. If $T_r$ is the time until $r$ arrivals,

  $$E[T_r] = \frac{1}{\lambda} + E[T_{r-1}],$$

  giving

  $$\boxed{E[T_r] = \frac{r}{\lambda}}.$$

- **The ranking/order-statistic example.** Here the remaining problem isn't identical, but it has the same structure with one fewer man:

  $$E_{m,w} = 1 + \frac{m}{m+w}E_{m-1,w},$$

  producing

  $$\boxed{E_{m,w} = \frac{m+w+1}{w+1}}.$$

More generally, this technique is extremely important for Markov chains, random walks, queueing systems, branching processes, reliability models, and stochastic dynamic programming. For example, if $T_i$ is the expected time to reach a target starting from Markov-chain state $i$, then

$$\boxed{T_i = 1 + \sum_j P_{ij}T_j}$$

for a non-target state $i$.

So there is a common principle behind all of these:

$$\boxed{\text{Expected total} = \text{expected contribution now} + \text{expected contribution later.}}$$

This is often called **first-step analysis**. Mathematically, it is simply the law of iterated expectation applied after conditioning on what happens first.

The geometric distribution is probably the cleanest next example to study because it gives exactly the form you have been using:

$$E[X] = 1 + \text{probability of continuing} \times E[X].$$

Then negative binomial shows how this generalizes when the state changes after a success.

### 1.6.3 Ranking example: deriving a PMF from the chain rule

We can also answer the ranking example by deriving PMF using the chain rule. Two men and two women receive distinct examination ranks, and every labeled ordering is equally likely. Let $X$ be the best (smallest numerical) rank achieved by a woman; the possible values are 1, 2, and 3, since rank 4 is impossible (only two men can occupy positions ahead of the first woman). Table 1.7 derives the PMF using the chain rule.

Table 1.7: Distribution of the best rank achieved by a woman

| $x$ | **Required gender pattern** | **Probability calculation** | $P(X = x)$ |
|---|---|---|---|
| 1 | W | 2 women / 4 people | 1/2 |
| 2 | M, then W | (2/4)(2/3) | 1/3 |
| 3 | M, M, then W | (2/4)(1/3)(2/2) | 1/6 |

For example, the chain rule gives the middle row explicitly:

$$P(X = 2) = P(G_1 = M)\, P(G_2 = W \mid G_1 = M) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

The PMF sums correctly, $1/2 + 1/3 + 1/6 = 1$, and the expected best rank is

$$E[X] = 1\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{3}\right) + 3\left(\tfrac{1}{6}\right) = \tfrac{5}{3} \approx 1.667,$$

a value that, like the family-size mean, is not itself an attainable rank but the average over repeated random rankings.

This calculation generalizes. With $n$ people including $w$ women, the set of positions occupied by women is uniformly distributed over the $\binom{n}{w}$ possible position sets. If $X$ is the best female rank, then for feasible $r$ the first woman occupies position $r$ and the remaining $w - 1$ female positions are chosen from the $n - r$ later positions:

$$\begin{aligned} P(X = r) &= \frac{\binom{n-r}{w-1}}{\binom{n}{w}}, \qquad r = 1, \ldots, n - w + 1, \\ E[X] &= \frac{n+1}{w+1}. \end{aligned} \tag{1.17}$$

For $n = 4$ and $w = 2$, the general formula gives $(4 + 1)/(2 + 1) = 5/3$, matching the direct calculation above.

### 1.6.4 Three-dice casino game

Three fair six-sided dice produce $6^3 = 216$ equally likely ordered outcomes. A bet on "Small" wins \$100 when the sum is 4 through 10, *except* that any triple is a house win; it loses \$100 for a "Big" sum of 11 through 17 and for all triples. There are six triples — $(1,1,1),\ldots,(6,6,6)$ — so $P(\text{triple}) = 6/216 = 1/36$. Removing triples leaves 210 outcomes, and dice symmetry maps every nontriple Small outcome to a nontriple Big outcome, so each class contains 105 outcomes:

$$\begin{aligned} P(\text{Small}) &= \frac{105}{216} = \frac{35}{72} \approx 0.4861, \\ P(\text{Loss}) &= \frac{35}{72} + \frac{1}{36} = \frac{37}{72} \approx 0.5139. \end{aligned} \tag{1.18}$$

Table 1.8 lays out the full payoff distribution for a \$100 Small bet.

Table 1.8: Casino payoff distribution for a \$100 Small bet

| **Event** | **Outcomes** | **Probability** | **Net payoff** $R$ |
|---|---|---|---|
| Small, nontriple | 105 | $35/72 = 0.4861$ | $+\$100$ |
| Big, nontriple | 105 | $35/72 = 0.4861$ | $-\$100$ |
| Triple | 6 | $1/36 = 0.0278$ | $-\$100$ |
| Total | 216 | 1.0000 | — |

The expected net return per \$100 bet is negative:

$$E[R] = 100\left(\frac{35}{72}\right) - 100\left(\frac{37}{72}\right) = -\frac{25}{9} \approx -\$2.78. \tag{1.19}$$

The player loses about \$2.78 per \$100 bet on average, so the house edge is approximately 2.78%; a single bet can still win, since expectation describes only the average over repeated play. By linearity, the expected cumulative return after $n$ independent bets is

$$E\left[\sum_{i=1}^{n} R_i\right] = nE[R] = -\frac{25n}{9}. \tag{1.20}$$

## 1.7 Variance, Standard Deviation, and Risk

### 1.7.1 Why the mean is not enough

Two distributions can share an expected value while carrying very different risk. Consider lottery $W$, which pays $-10$ or $+10$ with equal probability, and lottery $Y$, which pays $-100$ or $+100$ with equal probability:

$$E[W] = (-10)(0.5) + (10)(0.5) = 0, \qquad E[Y] = (-100)(0.5) + (100)(0.5) = 0.$$

Both means are zero, but $Y$ exposes the decision maker to gains and losses ten times larger than $W$'s. Preference between the two cannot be explained by expected value alone: a risk-averse person may prefer $W$, a risk-seeking person may prefer $Y$, and a

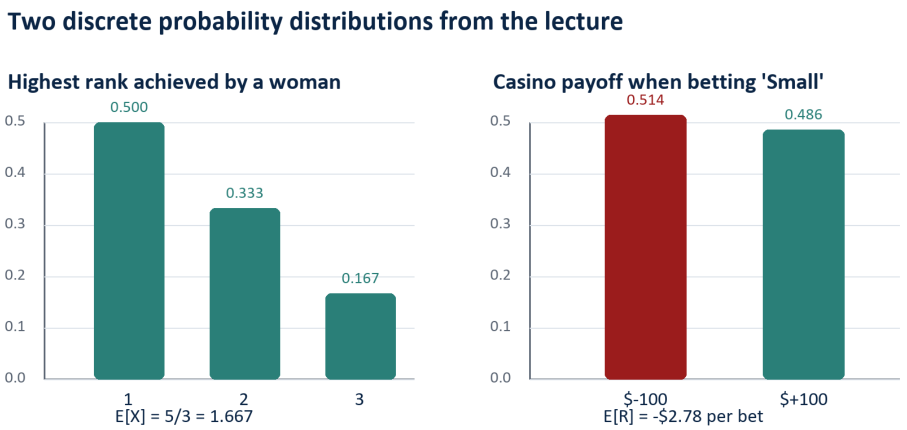


Figure 1.5: Probability mass determines both the center (expectation) and spread of a random variable.

complete decision model could apply expected utility, downside risk, value at risk, or another criterion,

$$\text{Expected utility} = \sum_x p(x)\, u(x), \qquad \text{where a concave } u \text{ represents risk aversion.}$$

### 1.7.2 Definitions

The expected deviation from the mean is always zero, $E[X - \mu] = E[X] - \mu = 0$, so positive and negative deviations cancel and a useful spread measure must remove their signs. **Variance** squares the deviations before averaging them:

$$\begin{aligned} \operatorname{Var}(X) = \sigma_X^2 &= E\Big[(X - \mu_X)^2\Big] \\ &= \sum_x (x - \mu_X)^2 p_X(x) \quad \text{or} \quad \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x)\, dx. \end{aligned} \tag{1.21}$$

Variance carries squared units — if $X$ is measured in dollars, variance is measured in dollars squared — so **standard deviation** returns to the original units and is easier to interpret:

$$SD(X) = \sigma_X = \sqrt{\operatorname{Var}(X)}. \tag{1.22}$$

Both variance and standard deviation are nonnegative and equal zero only when $X$ is constant with probability one. Table 1.9 summarizes $W$ and $Y$: both outcomes of $W$ are ten units from its zero mean, giving variance 100 and standard deviation 10; both outcomes of $Y$ are one hundred units from its zero mean, giving variance 10,000 and standard deviation 100, consistent with $Y = 10W$.

Table 1.9: Same expected value, different variability

| Game | Distribution | $E$[Game] | Variance | SD |
|---|---|---|---|---|
| $W$ | $+10$ or $-10$, each w.p. 1/2 | 0 | 100 | 10 |
| $Y$ | $+100$ or $-100$, each w.p. 1/2 | 0 | 10,000 | 100 |

$$SD(W) = \sqrt{\tfrac{1}{2}(10)^2 + \tfrac{1}{2}(-10)^2} = 10.$$

### 1.7.3 The computational identity and transformation rules

Expanding the square in Equation (1.21) gives an equivalent, often computationally convenient identity, although the centered formula is usually more numerically stable in software when the mean is very large relative to the spread:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2. \tag{1.23}$$

**Variance in Terms of Moments Expression**

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2.$$

This expression is verified as follows:

$$\begin{aligned}
\mathrm{Var}(X) &= \sum_x \big(x - E[X]\big)^2 p_X(x) \\
&= \sum_x \Big(x^2 - 2xE[X] + \big(E[X]\big)^2\Big) p_X(x) \\
&= \sum_x x^2 p_X(x) \; - \; 2E[X] \sum_x x p_X(x) \; + \; \big(E[X]\big)^2 \sum_x p_X(x) \\
&= E[X^2] - 2\big(E[X]\big)^2 + \big(E[X]\big)^2 \\
&= E[X^2] - \big(E[X]\big)^2.
\end{aligned}$$

**Example. Mean and Variance of the Bernoulli.** Consider the experiment of tossing a biased coin, which comes up a head with probability $p$ and a tail with probability $1-p$, and the Bernoulli random variable $X$ with PMF

$$p_X(k) = \begin{cases} p & \text{if } k = 1, \\ 1-p & \text{if } k = 0. \end{cases}$$

Its mean, second moment, and variance are given by the following calculations:

$$\begin{aligned}
E[X] &= 1 \cdot p + 0 \cdot (1-p) = p, \\
E[X^2] &= 1^2 \cdot p + 0 \cdot (1-p) = p, \\
\mathrm{Var}(X) &= E[X^2] - \big(E[X]\big)^2 = p - p^2 = p(1-p).
\end{aligned}$$

Shifting every outcome by a constant changes the mean but not the variance; scaling every outcome by $a$ multiplies variance by $a^2$ and standard deviation by $|a|$:

$$\begin{aligned} E[aX+b] &= aE[X]+b, \\ \operatorname{Var}(aX+b) &= a^2 \operatorname{Var}(X), \\ SD(aX+b) &= |a|\, SD(X). \end{aligned} \tag{1.24}$$

The absolute value matters because standard deviation cannot be negative: if $a = -2$, the distribution is reflected and stretched by two, so its standard deviation doubles rather than becoming negative.

As an illustration of the mean shifting under unequal probabilities, suppose the $Y$ lottery is modified so that $-100$ occurs with probability 0.4 and $+100$ with probability 0.6. The mean shifts to 20, since the positive outcome is now more likely, and the variance falls to 9,600:

$$\begin{aligned} E[Y] &= (-100)(0.4) + (100)(0.6) = 20, \\ \operatorname{Var}(Y) &= (-100-20)^2(0.4) + (100-20)^2(0.6) = 9{,}600, \\ SD(Y) &\approx 97.98. \end{aligned}$$

The drop from variance 10,000 to 9,600 reflects the shifted mean and unequal masses: the $+100$ outcome is now both closer to the mean and more likely, modestly reducing the average squared deviation.

For the casino payoff $R$ from Section 1.6, $R^2 = 10{,}000$ in both payoff states, so

$$\operatorname{Var}(R) = 10{,}000 - \left(-\tfrac{25}{9}\right)^2 \approx 9{,}992.28, \qquad SD(R) \approx \$99.96.$$

The expected loss is small relative to the volatility of a single bet, which helps explain why short winning streaks can coexist with a long-run house advantage.

### 1.7.4 Independence, covariance, and adding random variables

Variance is not linear the way expectation is. For *independent* random variables, the variance of a weighted sum is the sum of weighted variances, with each coefficient squared:

$$X_1 \perp X_2 \implies \operatorname{Var}(c_1X_1 + c_2X_2) = c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2). \tag{1.25}$$

For dependent variables, **covariance** captures how their deviations move together — positive covariance means above-average values of the two variables tend to occur together, negative covariance means one tends to be above average when the other is below:

$$\begin{aligned} \operatorname{Cov}(X_1, X_2) &= E[(X_1-\mu_1)(X_2-\mu_2)]\,, \\ \operatorname{Var}(c_1X_1 + c_2X_2) &= c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2) + 2c_1c_2 \operatorname{Cov}(X_1, X_2). \end{aligned} \tag{1.26}$$

Independence implies zero covariance when second moments exist, but zero covariance does not generally imply independence — a distinction that matters in regression, portfolio analysis, forecasting, and any system where multiple uncertain inputs move together.

The failure of the independent-sum formula is easy to see with $X_2 = 3X_1 + 5$: since $X_2$ is built directly from $X_1$, the two are perfectly dependent, and $\operatorname{Var}(X_2) = 9 \operatorname{Var}(X_1)$. Adding them gives $X_1 + X_2 = 4X_1 + 5$, so the exact variance is

$$\operatorname{Var}(X_1 + X_2) = \operatorname{Var}(4X_1 + 5) = 16 \operatorname{Var}(X_1),$$

sixteen times $\mathrm{Var}(X_1)$, not the ten times that the independent-sum formula would (incorrectly) suggest. The missing six units of variance come from twice the covariance: since $\mathrm{Cov}(X_1, 3X_1 + 5) = 3\,\mathrm{Var}(X_1)$,

$$2\,\mathrm{Cov}(X_1, X_2) = 2 \times 3\,\mathrm{Var}(X_1) = 6\,\mathrm{Var}(X_1).$$

Because covariance depends on measurement units, **correlation** is often used to express standardized dependence: dividing covariance by the two standard deviations produces a quantity between $-1$ and $+1$,

$$\mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{SD(X_1)\,SD(X_2)}, \qquad -1 \le \mathrm{Corr} \le 1. \tag{1.27}$$

Values near $+1$ indicate strong positive linear co-movement, values near $-1$ indicate strong negative linear co-movement, and values near zero indicate little linear association. For $X_2 = 3X_1 + 5$ with $X_1$ having positive variance, $\mathrm{Corr}(X_1, X_2) = +1$, since $X_2$ is a perfectly increasing affine function of $X_1$; a negative coefficient would instead give correlation $-1$. This makes visible, on a standardized scale, exactly the dependence that the naive independent-sum formula missed. The broader modeling lesson: whether risks diversify depends on their dependence structure. Adding independent risks can smooth relative uncertainty, while adding highly positively correlated risks provides little of that diversification benefit.

### 1.7.5 A counterexample: zero covariance without independence

The caveat raised in Lecture 1 is not a technicality that only matters for exotic distributions — it is easy to construct an explicit random variable pair with zero covariance that is nevertheless strongly dependent, in the strongest possible sense: one variable is a deterministic function of the other.

**1.7.5.0.1 A discrete example.** Let $X$ take the values $-1, 0, 1$, each with probability $1/3$, and define

$$Y = X^2.$$

Table 1.10 lays out the resulting joint distribution: since $Y$ is a deterministic function of $X$, only three $(x, y)$ pairs receive positive probability.

Table 1.10: Joint distribution of $X$ and $Y = X^2$

| $x$ | $y = x^2$ | $P(X = x, Y = y)$ |
|---|---|---|
| $-1$ | 1 | 1/3 |
| 0 | 0 | 1/3 |
| 1 | 1 | 1/3 |

**Zero covariance.** By symmetry, $E[X] = \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0$. Since $x^3 = x$ for each of $-1, 0, 1$ in this example, $E[XY] = E[X^3] = E[X] = 0$ as well, so

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - (0)\left(\tfrac{2}{3}\right) = 0.$$

**Strong dependence.** Despite this, $Y$ is completely determined by $X$ — knowing $X$ leaves no uncertainty about $Y$ at all, which is about as dependent as two random variables can be. The joint table confirms this directly: $P(X=0, Y=0) = 1/3$, but the marginals give $P(X=0)P(Y=0) = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = 1/9 \neq 1/3$, so the independence factorization from Equation (**??**) fails. Covariance is blind to this dependence because it only measures *linear* co-movement: as $X$ moves away from 0 in either direction, $Y$ increases either way, and the positive and negative contributions to $E[XY]$ cancel exactly.

**1.7.5.0.2 A continuous analogue.** The same construction works with a continuous variable: let $X \sim \text{Uniform}(-1, 1)$ and again set $Y = X^2$. By the same odd/even symmetry argument, $E[X] = E[X^3] = 0$, so $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$. Yet $Y$ is again an exact deterministic function of $X$, so the two variables are as dependent as possible. In both cases, the lesson is the same one flagged in Lecture 1: $\text{Cov}(X, Y) = 0$ rules out *linear* association, not dependence in general — only for pairs of Bernoulli indicators does zero covariance upgrade to full independence.

### 1.7.6 Law of Total Variance

**Where does it come from?** The law of total variance comes directly from the definition of variance plus the law of iterated expectations. The key algebraic trick is to add and subtract the conditional mean.

Let

$$\mu = E[X].$$

By definition,

$$\text{Var}(X) = E[(X-\mu)^2].$$

Now introduce some conditioning information $G$. Write

$$m_G = E[X \mid G].$$

Add and subtract $m_G$:

$$X - \mu = \underbrace{X - m_G}_{\text{variation within } G} + \underbrace{m_G - \mu}_{\text{variation across } G}.$$

Now square:

$$(X-\mu)^2 = (X-m_G)^2 + (m_G-\mu)^2 + 2(X-m_G)(m_G-\mu).$$

Take expectations:

$$\text{Var}(X) = E[(X-m_G)^2] + E[(m_G-\mu)^2] + 2E[(X-m_G)(m_G-\mu)].$$

The important part is showing that the last, cross-product term is zero.

**Why does the cross term disappear?** Consider

$$E[(X-m_G)(m_G-\mu)].$$

Use iterated expectation by conditioning on $G$:

$$E[(X-m_G)(m_G-\mu)] = E\Big[E[(X-m_G)(m_G-\mu) \mid G]\Big].$$

Once we condition on $G$, $m_G - \mu$ is known, so we can pull it outside:

$$= E\Big[(m_G - \mu)\, E[X - m_G \mid G]\Big].$$

But

$$E[X - m_G \mid G] = E[X \mid G] - m_G.$$

Since

$$m_G = E[X \mid G],$$

we have

$$E[X - m_G \mid G] = 0.$$

Therefore,

$$\boxed{E[(X - m_G)(m_G - \mu)] = 0.}$$

This is the crucial result.

**What remains?** We therefore have

$$\mathrm{Var}(X) = E[(X - m_G)^2] + E[(m_G - \mu)^2].$$

Look at the first term. Conditional on $G$,

$$E[(X - m_G)^2 \mid G] = \mathrm{Var}(X \mid G).$$

Using iterated expectation,

$$E[(X - m_G)^2] = E[\mathrm{Var}(X \mid G)].$$

The second term is

$$E[(m_G - \mu)^2].$$

But $m_G = E[X \mid G]$, and

$$E[m_G] = E\Big[E[X \mid G]\Big] = E[X] = \mu.$$

Therefore,

$$E[(m_G - \mu)^2] = \mathrm{Var}(m_G) = \mathrm{Var}\Big(E[X \mid G]\Big).$$

Putting everything together,

$$\boxed{\mathrm{Var}(X) = E[\mathrm{Var}(X \mid G)] + \mathrm{Var}\Big(E[X \mid G]\Big).}$$

**The intuition.** This formula says that the total variation in $X$ has two sources:

$$\boxed{\text{Total variance} = \text{average within-group variance} + \text{between-group variance.}}$$

For example, suppose $G$ represents gender and $X$ represents height. There is variation in height within each gender, represented by

$$E[\mathrm{Var}(X \mid G)],$$

and there can also be variation because the conditional means differ across genders, represented by

$$\mathrm{Var}\Big(E[X \mid G]\Big).$$

The reason we can simply add these two components is that the within-group deviation

$$X - E[X \mid G]$$

is orthogonal to every function of $G$, including

$$E[X \mid G] - E[X].$$

That orthogonality is exactly why the cross term vanishes.

This is also closely related to the regression ideas discussed earlier: conditional expectation is an orthogonal projection, and the law of total variance is essentially a Pythagorean theorem for that projection.

**Can we calculate the variance of ranking example by using the same trick?**

**Variance of the Ranking Example via the Law of Total Variance.** The same recursive idea works particularly nicely for the variance if we combine it with the law of total variance.

Let

$$X_{m,w} = \text{rank of the first woman}$$

when there are $m$ men and $w$ women. We already found

$$\mu_{m,w} = E[X_{m,w}] = \frac{m+w+1}{w+1}.$$

Now let

$$V_{m,w} = \mathrm{Var}(X_{m,w}).$$

The key recursive observation is the same as before:

- if the first person is a woman, $X = 1$;
- if the first person is a man, then

$$X = 1 + X_{m-1,w}.$$

**1. Condition on the first person.** Let $G$ indicate whether the first person is male or female. The law of total variance says

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid G)] + \mathrm{Var}(E[X \mid G]).$$

The two cases have probabilities

$$P(M) = \frac{m}{m+w}, \qquad P(W) = \frac{w}{m+w}.$$

If the first person is a woman,

$$X = 1,$$

so

$$\mathrm{Var}(X \mid W) = 0.$$

If the first person is a man,

$$X = 1 + X_{m-1,w}.$$

Adding 1 does not change variance, so

$$\mathrm{Var}(X \mid M) = V_{m-1,w}.$$

Therefore,

$$E[\mathrm{Var}(X \mid G)] = \frac{m}{m+w} V_{m-1,w}.$$

**2. Variance of the conditional means.** The two conditional means are

$$E[X \mid W] = 1$$

and

$$E[X \mid M] = 1 + \mu_{m-1,w}.$$

So the difference between the two possible conditional means is simply

$$\mu_{m-1,w}.$$

A two-point random variable with probabilities $p$ and $1-p$, whose two values differ by $d$, has variance

$$p(1-p)d^2.$$

Therefore,

$$\mathrm{Var}(E[X \mid G]) = \frac{m}{m+w} \cdot \frac{w}{m+w} \, \mu_{m-1,w}^2.$$

We already know

$$\mu_{m-1,w} = \frac{m+w}{w+1}.$$

Thus,

$$\mathrm{Var}(E[X \mid G]) = \frac{mw}{(m+w)^2} \left( \frac{m+w}{w+1} \right)^2 = \frac{mw}{(w+1)^2}.$$

Hence we obtain the beautiful variance recursion

$$\boxed{V_{m,w} = \frac{m}{m+w} V_{m-1,w} + \frac{mw}{(w+1)^2}}$$

with boundary condition

$$\boxed{V_{0,w} = 0}.$$

**3. Solve the recursion.** Let's calculate a few cases.

For $m = 1$,

$$V_{1,w} = \frac{1}{w+1} V_{0,w} + \frac{w}{(w+1)^2},$$

so

$$V_{1,w} = \frac{w}{(w+1)^2}.$$

For $m = 2$,

$$V_{2,w} = \frac{2}{w+2} V_{1,w} + \frac{2w}{(w+1)^2}.$$

Substituting,

$$V_{2,w} = \frac{2w}{(w+2)(w+1)^2} + \frac{2w}{(w+1)^2}.$$

Combining terms,

$$V_{2,w} = \frac{2w(w+3)}{(w+1)^2(w+2)}.$$

Continuing this pattern suggests

$$\boxed{V_{m,w} = \frac{mw(m+w+1)}{(w+1)^2(w+2)}}.$$

And this can easily be verified by induction using the recursion.

**4. Our example: 2 men and 2 women.** Set

$$m = 2, \qquad w = 2.$$

Then

$$\mathrm{Var}(X) = \frac{(2)(2)(2+2+1)}{(2+1)^2(2+2)}.$$

Therefore,

$$\mathrm{Var}(X) = \frac{20}{36} = \boxed{\frac{5}{9}}.$$

So

$$\boxed{E[X] = \frac{5}{3}, \qquad \mathrm{Var}(X) = \frac{5}{9}.}$$

For this particular example, the recursion itself is very short:

$$V_{2,2} = \frac{1}{2}V_{1,2} + \frac{4}{9},$$

and

$$V_{1,2} = \frac{1}{3}V_{0,2} + \frac{2}{9} = \frac{2}{9}.$$

Therefore,

$$V_{2,2} = \frac{1}{2}\left(\frac{2}{9}\right) + \frac{4}{9} = \frac{1}{9} + \frac{4}{9} = \boxed{\frac{5}{9}}.$$

This is the variance analogue of the earlier recursive expectation trick.

**Question: can we use the same trick to calculate the variance of geometric and exponential distribution?**

**Answer: Yes.**

**Variance via First-Step Analysis: Geometric and Exponential.** For both distributions, we can use exactly the same first-step / iterated-expectation trick. For variance, the easiest route is to recursively calculate the second moment $E[X^2]$, then use

$$\mathrm{Var}(X) = E[X^2] - E[X]^2.$$

**1. Geometric distribution.** Let

$$X \sim \mathrm{Geom}(p),$$

where $X$ is the number of trials until the first success. We already know from first-step analysis that

$$E[X] = \frac{1}{p}.$$

Let

$$M = E[X^2].$$

Look at the first trial. With probability $p$, we succeed immediately:

$$X = 1.$$

With probability $1-p$, we fail. We have used one trial and essentially restart the same problem. Let $X'$ be an independent copy of $X$. Then

$$X = 1 + X'.$$

Therefore, conditionally,

$$X^2 = (1+X')^2 = 1 + 2X' + X'^2.$$

Now use iterated expectation:

$$M = p(1^2) + (1-p)E[(1+X')^2].$$

Thus

$$M = p + (1-p)\left(1 + 2E[X] + M\right).$$

Since

$$E[X] = \frac{1}{p},$$

we have

$$M = p + (1-p)\left(1 + \frac{2}{p} + M\right).$$

Notice that

$$p + (1-p) = 1,$$

so

$$M = 1 + \frac{2(1-p)}{p} + (1-p)M.$$

Move the recursive term to the left:

$$pM = 1 + \frac{2(1-p)}{p}.$$

Therefore,

$$M = \frac{1}{p} + \frac{2(1-p)}{p^2} = \frac{2-p}{p^2}.$$

So

$$E[X^2] = \frac{2-p}{p^2}.$$

Finally,

$$\mathrm{Var}(X) = E[X^2] - E[X]^2$$

gives

$$\mathrm{Var}(X) = \frac{2-p}{p^2} - \frac{1}{p^2}.$$

Hence

$$\boxed{\mathrm{Var}(X) = \frac{1-p}{p^2}}.$$

So the entire geometric calculation came from the recursion

$$\boxed{E[X^2] = p + (1-p)E[(1+X)^2].}$$

**2. Exponential distribution.** Now let

$$X \sim \mathrm{Exp}(\lambda).$$

We know

$$E[X] = \frac{1}{\lambda}.$$

Can we do essentially the same thing? Yes, but continuous time makes it slightly more subtle. The cleanest approach is to look at a very short interval $h$.

For small $h$,

$$P(X \le h) \approx \lambda h,$$

while

$$P(X > h) \approx 1 - \lambda h.$$

If the event has not occurred by $h$, memorylessness tells us that the remaining waiting time is again exponential. Thus,

$$X = h + X'$$

after survival through the first $h$, where

$$X' \sim \mathrm{Exp}(\lambda).$$

Let

$$M = E[X^2].$$

If the event happens during the tiny interval, its contribution to $E[X^2]$ is of order $h^3$, so it disappears after taking the limit. If it does not happen,

$$X^2 = (h + X')^2 = h^2 + 2hX' + X'^2.$$

Therefore, to first order in $h$,

$$M \approx (1 - \lambda h)\, E[(h + X')^2].$$

Substitute:

$$M \approx (1 - \lambda h)\left(h^2 + 2hE[X] + M\right).$$

Since

$$E[X] = \frac{1}{\lambda},$$

we get

$$M \approx (1-\lambda h)\left(h^2 + \frac{2h}{\lambda} + M\right).$$

Ignore terms of order $h^2$ and higher. Then

$$M \approx M + \frac{2h}{\lambda} - \lambda h M.$$

Cancel $M$:

$$0 = \frac{2h}{\lambda} - \lambda h M.$$

Divide by $h$:

$$0 = \frac{2}{\lambda} - \lambda M.$$

Therefore,

$$M = \frac{2}{\lambda^2}.$$

Thus

$$E[X^2] = \frac{2}{\lambda^2}.$$

Now,

$$\mathrm{Var}(X) = E[X^2] - E[X]^2$$

so

$$\mathrm{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2}.$$

Hence

$$\boxed{\mathrm{Var}(X) = \frac{1}{\lambda^2}.}$$

**The common pattern.** This shows something deeper about the two distributions. For geometric:

$$X = \begin{cases} 1, & \text{success}, \\ 1 + X', & \text{failure}. \end{cases}$$

For exponential, over an infinitesimal interval:

$$X \approx \begin{cases} \text{event occurs during } h, & \\ h + X', & \text{event does not occur}. \end{cases}$$

In both cases, if we survive the first step, the remaining random variable has exactly the same distribution as the original one. That memoryless/restart property is why both their means and variances can be derived through the same recursive conditional-expectation argument.

### 1.7.7 Population parameters and sample estimators

The formulas above describe true, population-level moments; with observed data, analysts instead estimate them from a sample. Capital letters $X_i$ denote random observations, lowercase $x_i$ denote realized values, and the sample mean and (unbiased) sample variance are

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2. \tag{1.28}$$

The $n-1$ denominator corrects the downward bias created by estimating $\mu$ with the same sample mean used inside the sum; this is why software may report slightly different standard deviations depending on whether a population or sample convention is selected. Table 1.11 summarizes the correspondence, including the standard error of the sample mean, which reappears when confidence intervals are introduced later in the course.

Table 1.11: Population parameters and sample estimators

| **Target** | **Population quantity** | **Sample estimator** |
|---|---|---|
| Center | $\mu = E[X]$ | $\bar{x} = (1/n)\sum_i x_i$ |
| Variance | $\sigma^2 = E[(X-\mu)^2]$ | $s^2 = [1/(n-1)]\sum_i (x_i - \bar{x})^2$ |
| Standard deviation | $\sigma = \sqrt{\sigma^2}$ | $s = \sqrt{s^2}$ |
| Uncertainty of the sample mean | $SD(\bar{x}) = \sigma/\sqrt{n}$ | Estimated by $s/\sqrt{n}$ |

**Why the $n-1$ correction, more precisely.** If the population mean $\mu$ were known, variance could be estimated directly by averaging $(X_i - \mu)^2$ using all $n$ observations freely. In practice $\mu$ is unknown and is replaced by $\bar{X}$, but this substitution is not free: once $\bar{X}$ is fixed, the $n$ deviations $X_i - \bar{X}$ are constrained to sum to zero,

$$\sum_{i=1}^{n}(X_i - \bar{X}) = 0,$$

so only $n-1$ of them can vary independently — the last is always determined by the other $n-1$. One **degree of freedom** has been spent estimating $\bar{X}$, and dividing by $n$ instead of $n-1$ would systematically understate the population variance; the $n-1$ correction removes exactly that downward bias. The same bookkeeping recurs, in a more elaborate form, whenever a dataset is used to estimate several parameters that then enter a further calculation — for instance in regression, where each additional fitted coefficient consumes one more degree of freedom.

**Diminishing returns to more data.** Because $SD(\bar{X}) = \sigma/\sqrt{n}$, statistical error shrinks with sample size only at the rate $1/\sqrt{n}$. Quadrupling $n$ halves the standard error; to quarter it requires sixteen times as much data. Equivalently, cutting a target standard error in half is never achieved by doubling the sample — it requires roughly four times as much data. This square-root law is why moving from a mediocre estimate to a good one is often cheap, while moving from a good estimate to an excellent one can demand disproportionately more data; the same pattern reappears in machine learning, where reducing model error further and further can require very large increases in training data.

## 1.8 The Normal Distribution, Standardization, and Z Probabilities

### 1.8.1 The normal density

Unlike the uniform examples above, which have finite lower and upper bounds, a normal random variable has support across the entire real line. Its density is symmetric, peaks at the mean $\mu$, and decays rapidly as $x$ moves away from $\mu$; the parameter $\sigma$ is the standard deviation and $\sigma^2$ is the variance:

$$\begin{aligned} X &\sim N(\mu, \sigma^2), \\ f_X(x) &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad -\infty < x < \infty. \end{aligned} \tag{1.29}$$

Squaring $x - \mu$ makes the curve symmetric around $\mu$; the negative sign causes density to decrease as squared distance grows; and the factor $1/(\sigma\sqrt{2\pi})$ normalizes the total area to one. Changing $\mu$ shifts the entire curve without changing its shape; increasing $\sigma$ spreads the same total probability over a wider range and lowers the peak, while decreasing $\sigma$ concentrates probability and raises the peak. Symmetry implies that mean, median, and mode all coincide at $\mu$ for every normal distribution.

### 1.8.2 The empirical rule and tail behavior

Although normal distributions are unbounded, most probability mass lies within a few standard deviations of the mean. The canonical 68–95–99.7 empirical rule gives

$$P(\mu - \sigma \le X \le \mu + \sigma) \approx 0.6827, \qquad P(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 0.9545$$

and

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 0.9973.$$

The probability outside $\pm 3\sigma$ is therefore about 0.0027 in total, or roughly 0.00135 in each tail:

$$P(|X - \mu| > 3\sigma) \approx 1 - 0.9973 = 0.0027.$$

Rare does not mean impossible: the normal model still assigns positive probability to arbitrarily extreme finite observations.

### 1.8.3 Affine transformations and standardization

Normal distributions are closed under affine transformations: if $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then $Y$ is also normal, with its mean transformed linearly and its variance multiplied by $a^2$:

$$\begin{aligned} aX + b &\sim N(a\mu + b,\ a^2\sigma^2), \\ E[aX + b] &= a\mu + b, \\ SD(aX + b) &= |a|\sigma. \end{aligned} \tag{1.30}$$

A pure shift $X + b$ changes only the mean; a pure scale $aX$ changes both mean and spread. **Standardization** converts any nondegenerate normal variable to the standard normal distribution by subtracting $\mu$ (shifting the mean to zero) and dividing by $\sigma$ (rescaling the standard deviation to one):

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1). \tag{1.31}$$

The standard normal cumulative distribution function, written $\Phi$, returns the left-tail probability, and a Z table is simply a printed lookup table for $\Phi$:

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt, \qquad \Phi(-z) = 1 - \Phi(z).$$

### 1.8.4 Worked Z-table calculations

Suppose $X$ is normal with mean 100 and standard deviation 50. To find $P(X \le 110)$, standardize the cutoff:

$$z = \frac{110 - 100}{50} = 0.20, \qquad P(X \le 110) = \Phi(0.20) \approx 0.57926,$$

so about 57.9% of this distribution lies at or below 110. For an interval probability, standardize both endpoints and subtract cumulative left-tail areas; the interval from 90 to 110 maps to $-0.20$ through $+0.20$:

$$P(90 \le X \le 110) = \Phi(0.20) - \Phi(-0.20) = 0.57926 - 0.42074 = 0.15852.$$

The same method handles any interval: for a right-tail probability, subtract a left-tail table value from one; for a two-sided probability, standardize both bounds and subtract:

$$\begin{aligned} P(X > c) &= 1 - \Phi\left(\frac{c - \mu}{\sigma}\right), \\ P(a \le X \le b) &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right). \end{aligned}$$

It is important to always state whether a normal distribution's second parameter denotes variance or standard deviation; these notes write $N(\mu, \sigma^2)$, so the standard deviation used in a Z score is $\sigma$. Standardization converts a problem with arbitrary units into a universal reference scale: once the distribution of a standardized statistic is known, table values or software translate distances from the mean into probabilities, tail areas, critical values, and confidence levels — exactly the machinery that confidence intervals and hypothesis tests build on in later chapters.

### 1.8.5 Why is the normal distribution important? The Central Limit Theorem

A natural question is why the normal distribution deserves this much attention in the first place, rather than the uniform, triangular, or any other shape already introduced. Part of the answer lies in the behavior of the plug-in estimator from Section 1.4.1: the sample mean $\overline{X}$.

Because $\overline{X}$ is built from a random sample, it is itself a random variable with its own probability distribution — the **sampling distribution** of the sample mean. This is a different object from the population distribution of a single observation $X$: the population distribution describes individual outcomes, while the sampling distribution describes how the statistic $\overline{X}$ would vary if the sampling procedure were repeated many times.

Two properties of this sampling distribution follow directly from the expectation and variance rules developed earlier in this chapter. Since $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and expectation is linear,

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu, \tag{1.32}$$

so the sample mean is centered exactly at the population mean — the mathematical reason $\overline{X}$ is a sensible estimator of $\mu$ in the first place. Assuming the observations are independent, the variance of a sum is the sum of variances, and the constant $1/n$ enters squared:

$$\operatorname{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \operatorname{Var}(X_i) = \frac{1}{n^2}\left(n\sigma^2\right) = \frac{\sigma^2}{n}, \qquad SD(\overline{X}) = \frac{\sigma}{\sqrt{n}}. \tag{1.33}$$

This last quantity is exactly the standard error of the mean already listed in Table 1.11.

Equations (1.32) and (1.33) pin down the center and spread of $\overline{X}$, but not its *shape*. The **Central Limit Theorem** supplies that missing piece: as the sample size $n$ grows, the sampling distribution of $\overline{X}$ approaches a normal distribution,

$$\overline{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right), \tag{1.34}$$

*regardless of the shape of the original population distribution.* The population being averaged may be uniform, sharply right-skewed like the MBA salary data, or any other shape entirely — the sample mean nevertheless comes to resemble a bell curve as $n$ increases. This is why the normal distribution earns its central place in this chapter: it is not merely one convenient shape among many, but the shape that plug-in estimators such as $\overline{X}$ converge toward under very general conditions, which is what ultimately licenses the standardization and Z-table machinery developed in the rest of this section.

Standardizing $\overline{X}$ using Equation (1.34) gives a statistic with a known, tabulated distribution even when the population itself is unknown:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approx.}}{\sim} N(0, 1), \qquad \text{so} \qquad P\left(-1.96 \le \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \le 1.96\right) \approx 0.95. \tag{1.35}$$

This probability statement is the bridge from the Central Limit Theorem to confidence intervals: rearranging it so that the unknown $\mu$, rather than the observed $\overline{X}$, sits in the middle of the inequality produces $\overline{X} \pm 1.96\,\sigma/\sqrt{n}$, a 95% confidence interval for the population mean — the subject taken up in later chapters.

## 1.9 Interpretation Cautions for Research

Table 1.12 lists several informal shortcuts that are common in practice, why each is incomplete, and the more rigorous correction.

Table 1.12: Common shortcuts and the rigorous correction

| Informal shortcut | Why it is incomplete | Rigorous correction |
|---|---|---|
| A larger mean proves an effect. | Observed means vary across samples. | Evaluate the difference relative to its standard error and design. |
| The histogram is the probability density. | Bars depend on the sample and the chosen bins. | Treat it as an empirical estimate; report the bin construction. |
| A bell-shaped plot is normal. | Many distributions look approximately bell-shaped. | Use diagnostics and substantive modeling assumptions. |
| A numeric category code has quantitative meaning. | Codes may be arbitrary labels. | Model categories explicitly unless the ordering is substantive. |
| Equal means imply equivalent populations. | Spread and shape may differ strongly. | Compare variability, distributional shape, and the target estimand. |

**Statistical-significance clarification.** A difference between treatment and control means is descriptive evidence, not automatically a significant effect. Significance depends on the estimated difference, group variability, sample sizes, the assignment mechanism, the dependence structure, and the null model used for inference.

## 1.10 Compact Formula Reference

Table 1.13: Core formulas from this lecture

| Concept | Formula | Purpose |
|---|---|---|
| Random variable | $X : \Omega \to \mathbb{R}$ | Map outcomes to numerical values |
| Discrete PMF | $p_X(x) = P(X = x)$; $\sum_x p_X(x) = 1$ | Allocate mass across discrete values |
| Continuous PDF | $\int_a^b f_X(x)\,dx = P(a \le X \le b)$ | Probability as area under a density |
| CDF | $F_X(x) = P(X \le x)$ | Accumulate probability through $x$ |
| Uniform density | $f_X(x) = 1/(b-a)$ for $a \le x \le b$ | Constant density on a bounded interval |
| Empirical bin mass | $\widehat{p}_j = n_j/N$ | Convert histogram counts to proportions |
| Discrete expectation | $E[X] = \sum_x x\,p_X(x)$ | Probability-weighted center |
| Continuous expectation | $E[X] = \int x f_X(x)\,dx$ | Density-weighted center |

Table 1.13: Core formulas from this lecture

| Concept | Formula | Purpose |
|---|---|---|
| Linearity | $E[aX + b] = aE[X] + b$ | Transform means without re-deriving distributions |
| Variance | $\mathrm{Var}(X) = E[(X - \mu)^2]$ | Average squared distance from the mean |
| Variance identity | $\mathrm{Var}(X) = E[X^2] - (E[X])^2$ | Efficient computation |
| Affine variance rule | $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$ | Rescale variance under a linear transform |
| Standard deviation | $SD(X) = \sqrt{\mathrm{Var}(X)}$ | Spread in original units |
| Covariance | $\mathrm{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$ | Co-movement of two variables |
| Variance of a sum | $\mathrm{Var}(c_1X_1 + c_2X_2) = c_1^2 \mathrm{Var}(X_1) + c_2^2 \mathrm{Var}(X_2) + 2c_1c_2 \mathrm{Cov}(X_1, X_2)$ | Combine dependent risks |
| Correlation | $\mathrm{Corr}(X_1, X_2) = \mathrm{Cov}(X_1, X_2)/[SD(X_1)SD(X_2)]$ | Standardized, unit-free dependence |
| Normal density | $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp[-(x - \mu)^2/(2\sigma^2)]$ | Bell-shaped continuous model |
| Standardization | $Z = (X - \mu)/\sigma$ | Convert to the $N(0, 1)$ reference scale |
| Interval probability | $P(a \leq X \leq b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$ | Normal probabilities via the Z table |
| Sample mean | $\bar{X} = (1/n) \sum_i X_i$ | Estimate the population mean |
| Sample variance | $s^2 = [1/(n - 1)] \sum_i (X_i - \bar{X})^2$ | Estimate the population variance |

## 1.11 Practice Problems with Concise Solutions

### 1.11.1 Coding versus meaning

A dataset codes departments as 0, 1, and 2. Is the resulting variable quantitative?

**Solution.** Not necessarily. If the codes only identify categories, arithmetic comparisons such as "department 2 is twice department 1" are meaningless. The variable is categorical even though it is stored numerically.

### 1.11.2 Empirical bin probability

A salary bin contains 519 of 2,597 observations. Estimate its probability mass.

**Solution.** $\hat{p} = 519/2{,}597 \approx 0.200$, so the bin represents about 20.0% of the observed population.

### 1.11.3 Ranking distribution

Verify that the probabilities for $X$, the best female rank among two men and two women, form a valid PMF.

**Solution.** Each probability is nonnegative, and $1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 1$.

### 1.11.4 Casino expectation

A Small bet wins \$100 with probability 35/72 and loses \$100 with probability 37/72. Find the expected payoff.

**Solution.** $E[R] = 100(35/72) - 100(37/72) = -200/72 = -25/9 \approx -\$2.78$.

### 1.11.5 Variance from a PMF

Let $Z$ equal $-2$ with probability 0.25 and 2 with probability 0.75. Find $E[Z]$, $\mathrm{Var}(Z)$, and $SD(Z)$.

**Solution.** $E[Z] = -2(0.25) + 2(0.75) = 1$. $E[Z^2] = 4$, so $\mathrm{Var}(Z) = 4 - 1^2 = 3$ and $SD(Z) = \sqrt{3} \approx 1.732$.

### 1.11.6 Same mean, different risk

Why are the $\pm 10$ and $\pm 100$ games not equivalent even though both expected values are zero?

**Solution.** Their standard deviations are 10 and 100. The second game produces outcomes ten times farther from the common mean and therefore carries substantially greater risk.

## 1.12 Chapter Summary

The chapter follows a single pipeline: define a numerical random variable and its support; specify a normalized PMF or PDF; distinguish theoretical probabilities from finite-sample empirical frequencies; use transformations to connect variables and derive their means and variances, including for sums of dependent variables; and, when a normal model is appropriate, standardize to $Z$ and calculate areas with the standard normal distribution.

Model $\to$ distribution $\to$ sample $\to$ summary measures $\to$ standardized inference.

The generative-AI examples fit the first half of this pipeline: model outputs are random variables, softmax supplies normalized conditional masses, and decoding samples from those masses. The statistical examples fit the second half: expectation summarizes center, variance summarizes spread, covariance captures dependence, and standardization makes normal probabilities comparable across measurement scales. Random variables are, in this sense, the bridge between a qualitative sample space and numerical statistical analysis: their distributions answer which values are possible and how likely those values are, histograms provide an empirical view of a distribution, expectation compresses location into a single balance point, and variance and standard deviation describe the dispersion that the mean alone cannot capture. These same population quantities become sample estimators, standard errors, and formal comparisons once the population distribution is unknown — the subject of the chapters that follow.

### Rapid Review

- A random variable maps experimental outcomes to numbers; it may be discrete or continuous.
- A PMF sums to one; a PDF integrates to one, and probability is area under the density.
- For Uniform$(a, b)$, density is $1/(b-a)$; a histogram is an empirical estimate of a distribution, not the distribution itself.
- An affine transformation $Y = aX + b$ shifts and rescales a distribution; the Jacobian rule extends this to nonlinear transformations.
- The mean is a probability-weighted balance point; expectation is linear, even for dependent variables.
- Variance is the expected squared distance from the mean; standard deviation is its square root and shares the original units.
- Scaling by $a$ multiplies variance by $a^2$; shifting by $b$ does not change variance.
- Variances add directly only under zero covariance; independence is sufficient but not necessary for that.
- A normal variable has bell-shaped density parameterized by $\mu$ and $\sigma^2$; standardization $Z = (X - \mu)/\sigma$ converts normal probabilities to the $N(0, 1)$ reference scale.
- Population parameters ($\mu$, $\sigma^2$) are estimated from data by sample statistics ($\bar{x}$, $s^2$), with the sample variance's $n - 1$ denominator correcting for using an estimated mean.

## Appendix: Minimal Reproducibility Checklist

1. Confirm the file name, sheet name, column names, and number of nonmissing observations.
2. Inspect minimum and maximum values before choosing bin edges.
3. Check that every nonmissing value received a bin and that counts sum to $N$.
4. Check that normalized frequencies sum to 1 up to rounding.
5. Calculate the mean and standard deviation from raw values; use grouped calculations only when necessary.
6. Save the code and bin definitions with the analysis so the figure is reproducible.

# References